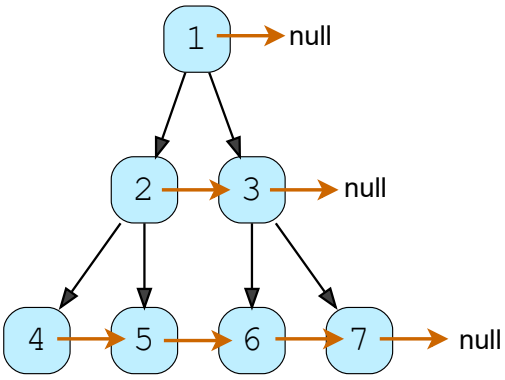
Solution: Connect Level Order Siblings

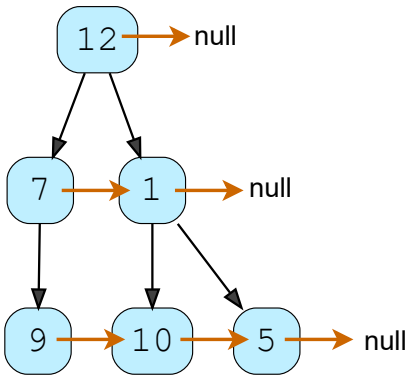
Problem Statement

Given a binary tree, connect each node with its level order successor. The last node of each level should point to a `null` node.

Example 1:



Example 2:

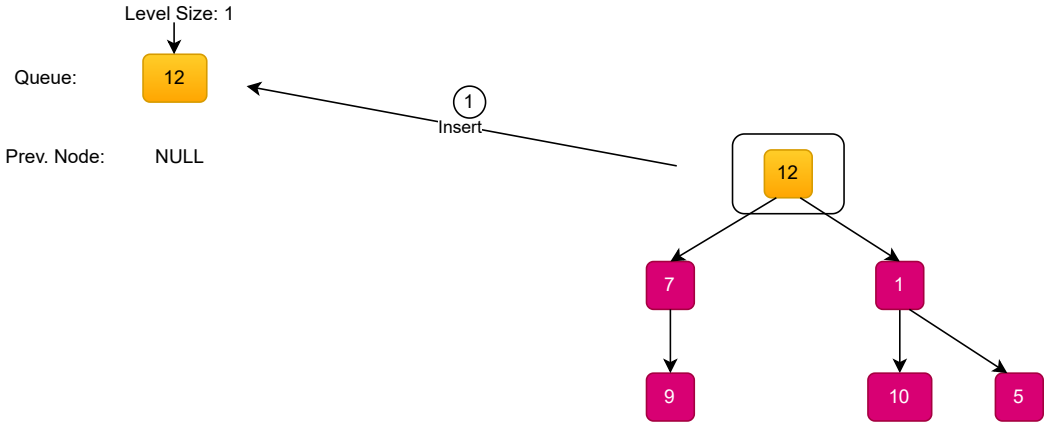


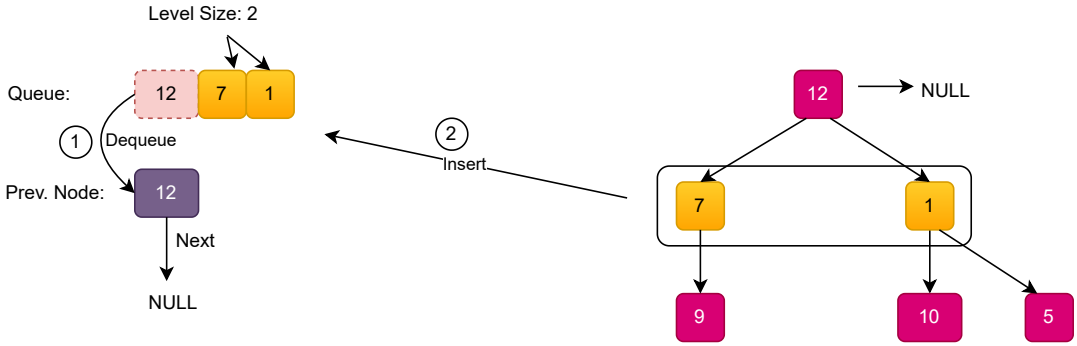
Solution

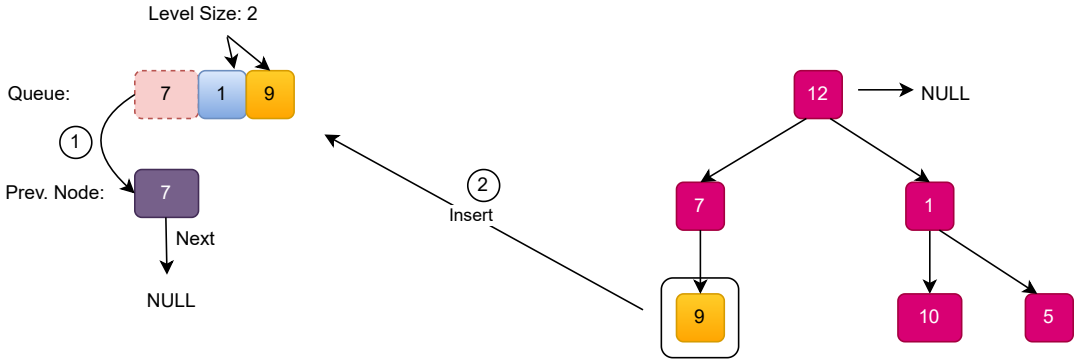
This problem follows the Binary Tree Level Order Traversal pattern. We can follow the same BFS approach. The only difference is that while traversing a level we will remember the previous node to

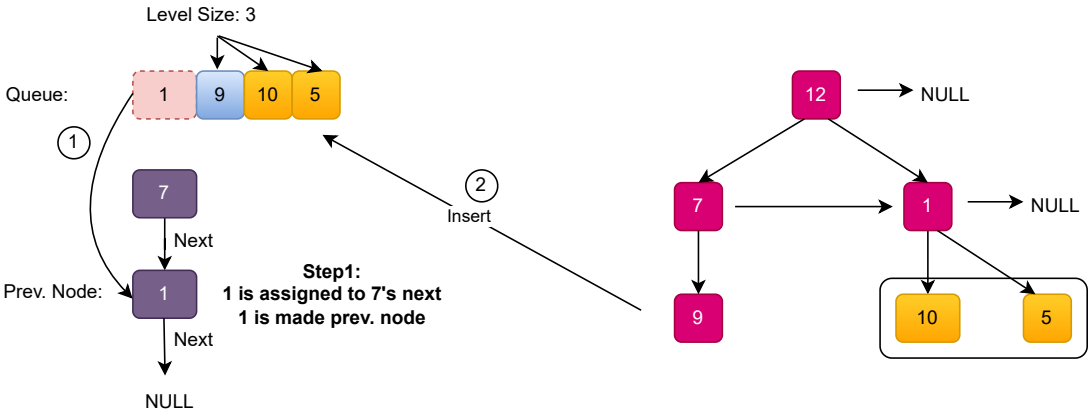
connect it with the current node.

Here is the visual representation of the algorithm:

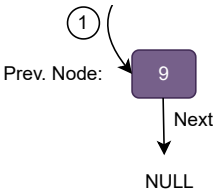

Level Size: 1
Queue:
12
Prev. Node:
NULL
1
Insert
12
7
1
9
10
5

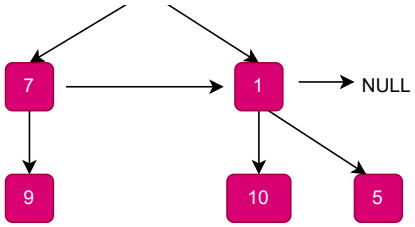

Level Size: 2
Queue:
12
7
1
1
Dequeue
Prev. Node:
12
Next
NULL
2
Insert
12
NULL
7
1
9
10
5

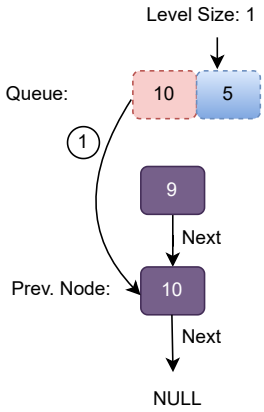

Level Size: 2
Queue:
7
1
9
1
Prev. Node:
7
Next
NULL
2
Insert
12
NULL
7
1
9
10
5

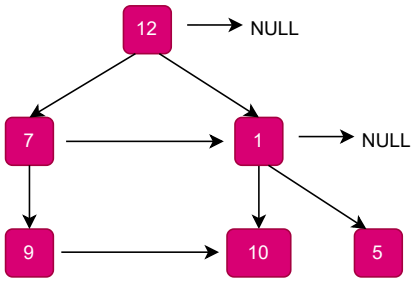

Level Size: 3
Queue:
1
9
10
5
1
7
Next
Prev. Node:
1
Next
NULL
Step1:
1 is assigned to 7's next
1 is made prev. node
2
Insert
12
NULL
7
1
NULL
9
10
5

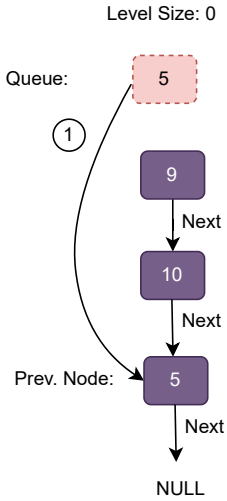

Level Size: 2
Queue:
9
10
5
12
NULL

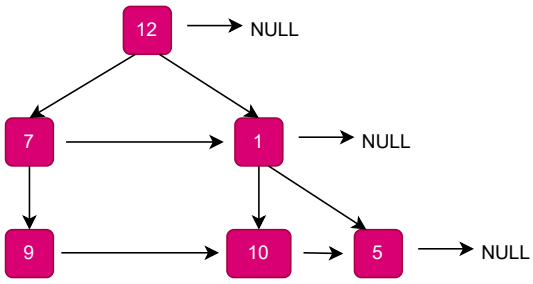


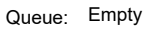


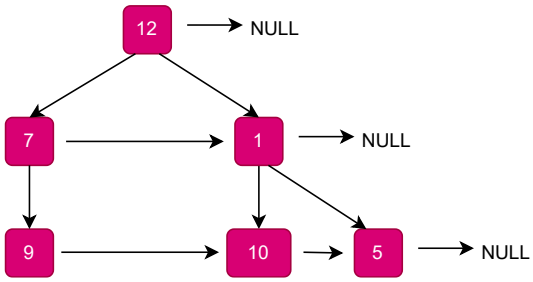








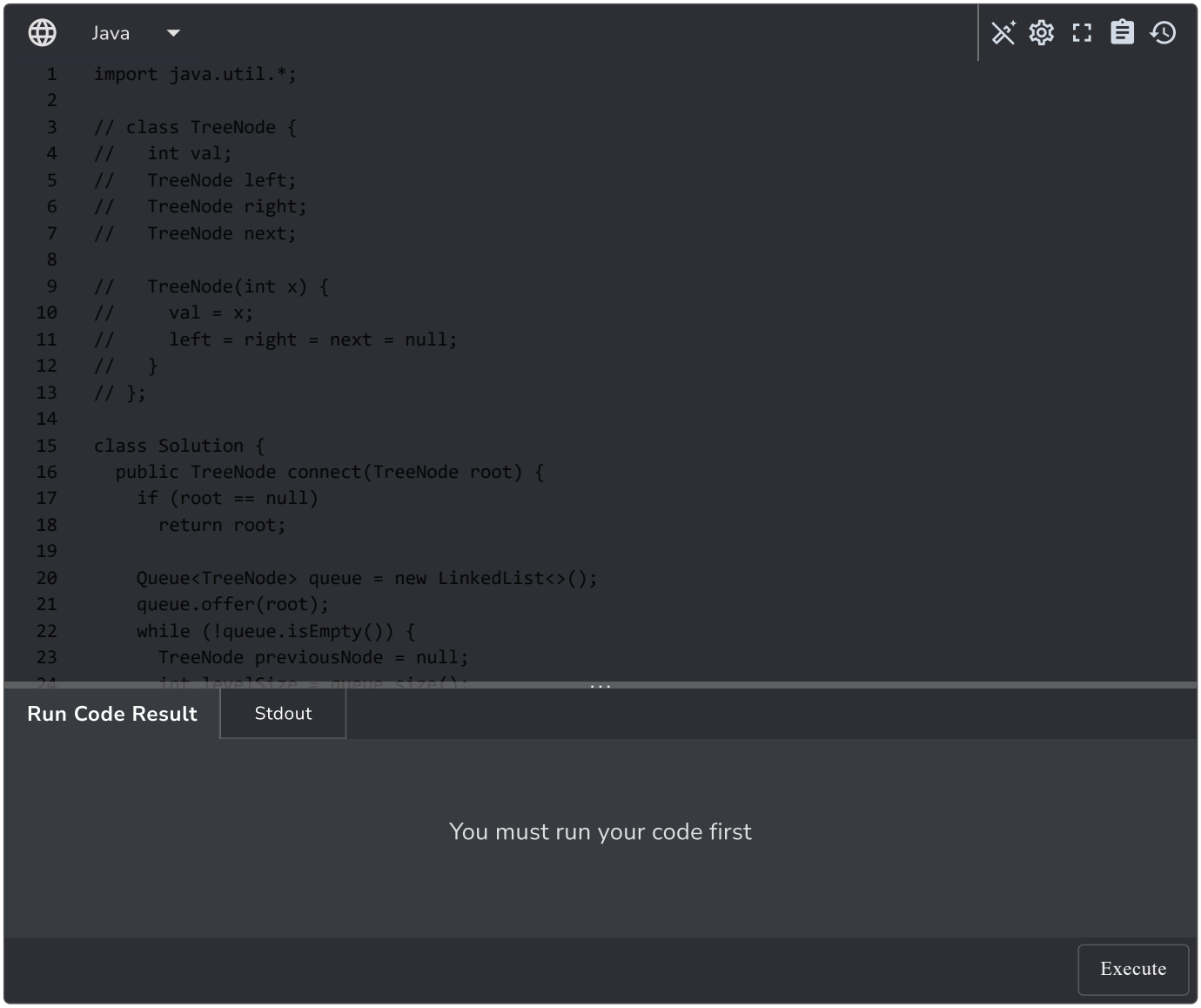




Connect Level Order Siblings

Code

Here is the code for this algorithm:



Time Complexity

The time complexity of the above algorithm is $O(N)$, where 'N' is the total number of nodes in the tree. This is due to the fact that we traverse each node once.

Space Complexity

The space complexity of the above algorithm will be $O(N)$, which is required for the queue. Since we can have a maximum of $N/2$ nodes at any level (this could happen only at the lowest level), therefore we will need $O(N)$ space to store them in the queue.

← Previous

Connect Level Order Siblings (Medium)

Next →

Problem Challenge 1: Connect All Level Order Siblings (Medium)